```
export default function cartReducer(state: StateType, action: ActionType) {
    switch (action.type) {
        case 'ADD_TO_CART':
            return {
                cartItems: [...state.cartItems, action.payload],
                total: state.total + action.payload.amount,
                quantity: state.quantity + 1
            }
        case 'INCREMENT': let cartItems = state.cartItems;
            cartItems.forEach(item => {
                if (item.id === action.payload) {
                    item.qty++;
                    item.amount = item.qty * item.price;
                }
            });
            return {
                cartItems,
                total: state.cartItems.map(item => item.amount).reduce((v1, v2) => v1 + v2, 0.0),
                quantity: state.quantity
            }
        case 'CLEAR_CART': return { cartItems: [], total: 0, quantity: 0 }
        default: return state;
    }
}
```

```
const initialState = {
    cartItems: [],
    total: 0,
    quantity: 0
}

export default function CartProvider({ children }: AppProps) {
    let navigate = useNavigate();
    let [state, dispatch] = useReducer(cartReducer, initialState);
    function addToCart(item: Cart) {
        dispatch({ type: 'ADD_TO_CART', payload: item });
    }

    function increment(id: number) {
        dispatch({ type: 'INCREMENT', payload: id })
    }
    function checkout() {
        // axios.post...
        dispatch({ type: 'CLEAR_CART' });
        navigate('/');
    }
    return <CartContext.Provider value={
        {
            cartItems: state.cartItems, total: state.total, quantity: state.quantity,
            addToCart,
```

1

cartItems: [],
total: 0,
quantity: 0

HEAP
State

2A

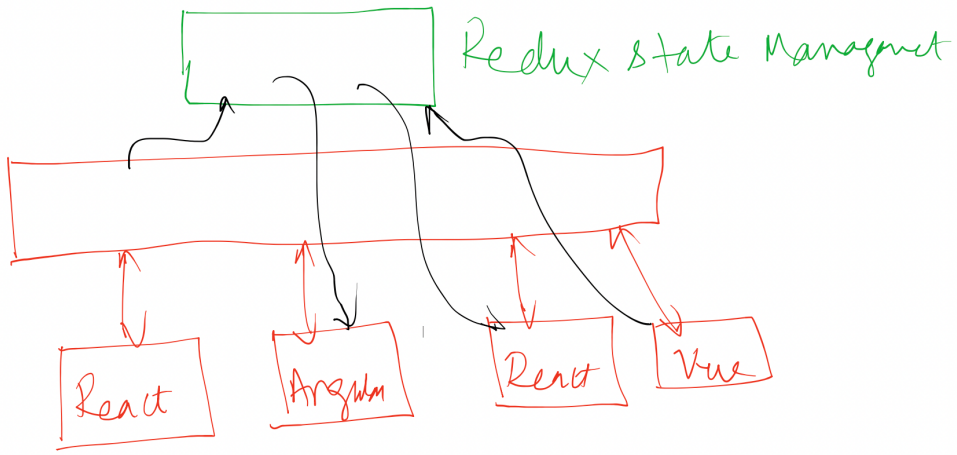
2B

3

cartItems: [1,
total: 8,
quantity: 8

State

4

Context

MicroFrontend